CRITERION FOR AN ACTION QUOTIENT STACK TO BE ALGEBRAIC

THE STACKS PROJECT AUTHORS

Editorial note. The closing 'Details omitted.' covers the isomorphism between the two displayed morphism descriptions via the inverse of G; retained verbatim and flagged for the reviewer. The condition-(2) identification is the tautological restriction of the groupoid.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 3428 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- General quotient criterion. lemma-quotient-algebraic (same chapter, named). The general groupoid criterion being specialized.
- Action groupoid. spaces-groupoids-lemma-groupoid-from-action (cross-chapter, named). The associated groupoid (X, G x_B X, s, t, c).

Source reference. Original lemma 06PJ: sources/stacks-criteria/pinned-original-source.tex, source lines 3305–3305. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 044Q: focused-seeds/geometry/S9-crit-group-quotient-algebraic/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0444: focused-seeds/geometry/S9-crit-group-quotient-algebraic/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme and let B be an algebraic space over S. Let G be a group algebraic space over B. Let X be an algebraic space over B and let $a : G \times_B X \to X$ be an action of G on X over B. The quotient stack $[X/G]$ is an algebraic stack if and only if there exists a morphism of algebraic spaces $\varphi : X' \to X$ such that*

(1) *$G \times_B X' \to X$, $(g, x') \mapsto a(g, \varphi(x'))$ is a surjection of sheaves, and*

Date: stacks-project:criteria.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(2) *the two projections $X'' \to X'$ of the algebraic space X'' given by the rule*
$$T \longmapsto \{(x'_1, g, x'_2) \in (X' \times_B G \times_B X')(T) \mid \varphi(x'_1) = a(g, \varphi(x'_2))\}$$
are flat and locally of finite presentation.

Proof. This lemma is a special case of Lemma 06PJ. Namely, the quotient stack $[X/G]$ is by Groupoids in Spaces, Definition 044Q equal to the quotient stack $[X/G \times_B X]$ of the groupoid in algebraic spaces $(X, G \times_B X, s, t, c)$ associated to the group action in Groupoids in Spaces, Lemma 0444. There is one small observation that is needed to get condition (1). Namely, the morphism $s : G \times_B X \to X$ is the second projection and the morphism $t : G \times_B X \to X$ is the action morphism a. Hence the morphism $h : U' \times_{g,U,t} R \to R \xrightarrow{s} U$ from Lemma 06PJ corresponds to the morphism
$$X' \times_{\varphi,X,a} (G \times_B X) \xrightarrow{\mathrm{pr}_1} X$$
in the current setting. However, because of the symmetry given by the inverse of G this morphism is isomorphic to the morphism
$$(G \times_B X) \times_{\mathrm{pr}_1,X,\varphi} X' \xrightarrow{a} X$$
of the statement of the lemma. Details omitted. □